

# NVIDIA – INTERVIEW EXPERIENCE

## About the Company:

NVIDIA (short for *Invention of Visual Display Interface Accelerator*) is a global leader in graphics processing technology and artificial intelligence computing. Known for its powerful GPUs, NVIDIA revolutionized gaming, professional visualization, and high-performance computing. Beyond graphics, it plays a key role in advancing AI, deep learning, data centres, and autonomous vehicles through its CUDA platform and NVIDIA AI frameworks.

NVIDIA India, with offices in Bengaluru, Hyderabad, and Pune, serves as a major R&D and engineering hub, driving innovations in GPU architecture, AI research, and cloud technologies. The teams focus on developing next-generation computing solutions that power industries ranging from gaming to scientific research and enterprise AI.

**Role Offered:** System Software Engineer

## Selection Process:

- 1. Online Assessment**
- 2. Technical Interview**

The Online Assessment was conducted at CUIC on 25/09/2025. But the shortlisted candidates were announced on 13/10/2025, So if you are confident enough in your performance be patient.

## Online Assessment:

- Duration: 1 hour
- Platform: HackerRank
- Pattern: 2 coding questions (one easy and one hard) & 27 MCQs including C/C++ fundamentals, OS, DSA, Debug and Aptitude.
- For the aptitude questions each carries 3 points, for technical questions each carries 2 points, the easy question carries 10 or 15 points and the hard one carries 75 points.

The MCQs were split into two sections 15 aptitude questions and 12 technical questions. For the aptitude section focus on permutation, profit percent, Chain rule problem and basic math questions.

For the technical questions focus on OS concepts like scheduling algorithms, paging, C/C++ fundamentals like string literals, pointers, static variables and functions, bit manipulation, signed/unsigned char or int, and some DSA analyzes and debug questions.

For example,

```
unsigned int y = 2;
```

```
int x = -2;
```

```
if(x > y)
```

```
    printf("X is greater than Y");
```

```
else
```

```
    printf("Y is greater than X");
```

**Output:** X is greater than Y

This question was asked to me even in the technical interview.

```
unsigned char ch = 0;
```

```
int count = 0;
```

```
while(++ch <= 255)
```

```
    count++;
```

```
printf("%d", count);
```

**Output:** This forms an infinite loop.

For the DSA, the easy question should be solved only using C and the hard question can be solved using either C/C++ ( **C and C++ are the only programming languages available for the OA** ).

**The first question is powers of two where a binary char array is provided and you have to check whether it is a power of 2.**

( The logic is based on the mathematical property that a positive number is a power of two if and only if its binary representation contains **exactly one '1'** )

```
int isPowerOfTwo(const char *str) {
```

```
    int one_count = 0;
```

```
    int i = 0;
```

```
    while (str[i] != '\0') {
```

```
        if (str[i] == '1') {
```

```
            one_count++;
```

```
        }
```

```
        i++;
```

```
    }
```

```
    return (one_count == 1);
```

```
}
```

```
int main() {
```

```
    int n;
```

```
    int i;
```

```
    char binary_string[101];
```

```
    scanf("%d", &n);
```

```
    for (i = 0; i < n; i++) {
```

```
        scanf("%100s", binary_string);
```

```
if (isPowerOfTwo(binary_string)) {  
    printf("True\n");  
} else {  
    printf("False\n");  
}  
}  
return 0;  
}
```

The second question is a hard recursion based string question.

**Link:** <https://leetcode.com/problems/special-binary-string/description/>

**These both questions were repeated for the past two years** so it is important not only for NVIDIA but also for any companies to research about the pattern of the hiring and **previously asked question's patterns**.

You can find these patterns in platforms like **leetcode, gfg, and sometimes reddit**. ( <https://leetcode.com/discuss/post/5896082/nvidia-software-engineering-internship-6-ngsp/> )

### **Technical Interview:**

The experience may vary but I was asked one DSA question and everything else was about bit manipulation and C/C++ fundamentals.

**Note:** It is **very important to revise the questions that were asked during the OA** for this round, some questions were asked from the OA and even the interviewer mentioned this (like the question that I mentioned above).

The DSA question is a linked list problem,

Link: <https://leetcode.com/problems/reverse-nodes-in-k-group/description/>

And for the other technical questions I was asked

```
bool Dummy(unsigned int n)
{
    if(n <= 0)
        return false;
    else
        return !(n & (n-1));
}
```

So you have to provide an appropriate name for the given function. ( This is a function to find if the given number is a power of 2)

Are these valid or throws an error?

If( x == 10) **ans: (valid)**

If( 10 == x) **ans: (valid)**

If( x = 10) **ans: (valid)**

If( 10 = x) **ans: (Invalid)**

And finally I was asked how a float point number is represented and stored in binary inside computer memory. ( **Ans:** using **Sign bit, Exponent and Mantissa**)

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